

Calibrated Is Not Actionable: Physics-Guided Posterior Distillation for Methane Source Attribution

Eric Liang

Department of Computer Science, Boston University
eric.jinchen.liang@gmail.com

Abstract—Satellites now detect methane super-emitters routinely, yet only $\sim 1\%$ of alerts lead to repairs: the bottleneck is *attribution* — naming a source precisely enough to act, which U.S. EPA rules quantify as a 50 m radius. On a methane testbed whose closed-form physics makes the *exact* Bayesian posterior computable for every scenario, we show that a conformal-wrapped deep localizer with a verified 90% coverage guarantee still yields a median search radius of 94 m — honestly calibrated, but far above both the 50 m bar and the 7 m information limit the sensor data actually supports. We diagnose this gap by running the same conformal audit on every pipeline stage, acquitting the sensors (7 m) and the distillation teacher (11.7 m) and convicting a missing computation: set encoders cannot perform per-cell matched filtering internally, and no supervision — sharper targets, transport losses, or a modern flow-based posterior head — teaches what the forward pass cannot represent. Our recipe hands the computation over: two closed-form matched-filter maps as input channels, a zero-initialized residual head, and distillation from the tempered exact posterior. Across three architecture families and nine seed-runs, every model lands under the 50 m bar (43.6–49.3 m) at unchanged coverage and latency, outperforming a fairness-hardened flow-based neural posterior estimator (55.5–71.2 m). Under realistic wind-measurement error the exact-physics pipeline collapses outright (coverage 0.70) while the distilled students remain calibrated, and modeling the wind uncertainty in the input maps recovers a further 12 m.

Index Terms—methane monitoring, conformal prediction, simulation-based inference, knowledge distillation, sensor networks, physics-guided machine learning

I. INTRODUCTION

Methane delivers roughly $80\times$ the 20-year warming potency of CO_2 and decays within about a decade, making rapid repair of oil-and-gas *super-emitters* one of the highest-leverage near-term climate actions. Detection is increasingly solved: UNEP’s satellite alert system issued over a thousand notifications in its first two years — of which $\sim 1\%$ saw documented action [1]. The missing step is attribution: converting a detection into a source location trustworthy enough to dispatch a crew. The EPA Super-Emitter Program makes “trustworthy enough” concrete — within 50 m [2] — and EU methane regulation imposes analogous source-level duties [3]. We treat 50 m as an engineering target.

Continuous monitoring with sparse networks of fixed point sensors is where attribution must happen: a handful of cheap masts around a facility, always on, catching the plume as it sweeps across them under naturally meandering wind. A

neural localizer maps readings to a probability map over source locations, and split conformal prediction [4] converts that map into a region guaranteed to contain the source 90% of the time. But coverage is a statement about frequency, not usefulness: a region can be honestly 90% and still be too large to act on. This paper measures that gap exactly, diagnoses its mechanism by elimination, and closes it with a deliberately simple recipe.

Central idea. When training data come from a simulator, the exact Bayesian posterior of every training scenario is computable offline. We use it twice: as a *reference* that turns “how sharp could a calibrated region be” into a number, and as a *teacher*. The novelty is not any component — matched filters are 1940s detection theory, distillation and conformal prediction are standard — but the diagnosis-driven composition, and the finding it certifies: *the binding constraint on calibrated localizers is a computation that must be handed to the network, not a lesson that can be taught to it.*

II. TESTBED AND EXACT-POSTERIOR AUDIT

CH4-T places a source of unknown location and rate ($10\text{--}500\text{ kg h}^{-1}$, log-uniform, bracketing the 100 kg h^{-1} super-emitter threshold) on a $500\times 500\text{ m}$ site sensed by 4–12 masts (ppm readings, 0.2–3 ppm noise, dropout). The plume is a steady Gaussian under known Pasquill–Gifford stability with Briggs open-country dispersion (coefficients verified against published tables; plume mass flux conserved to 4×10^{-6}), driven quasi-statically by a known fluctuating wind record ($1\text{--}8\text{ m s}^{-1}$ with AR(1) meander; we interpret steps as 1-min averages, a 30-min window). As the wind veers, the plume sweeps the masts: time-of-arrival information reduces weakly-identifiable scenarios from 57% (constant wind) to 6%.

Because the response is closed-form and linear in rate, the exact posterior over a 64×64 grid is computable per scenario (adaptive rate quadrature; float64; normalization error $\sim 10^{-12}$). A blocking gate requires conformal prediction applied to the exact posterior itself to self-calibrate (0.95 observed) — a gate that caught two real implementation bugs during development. All models share identical data (10k/1k/2k/2k train/val/calib/test, at the modal simulation budget of SBI benchmarks [7]), identical D4 symmetry augmentation, and an identical randomized-HPD split-conformal wrapper at verified 90% coverage, computed in numerically

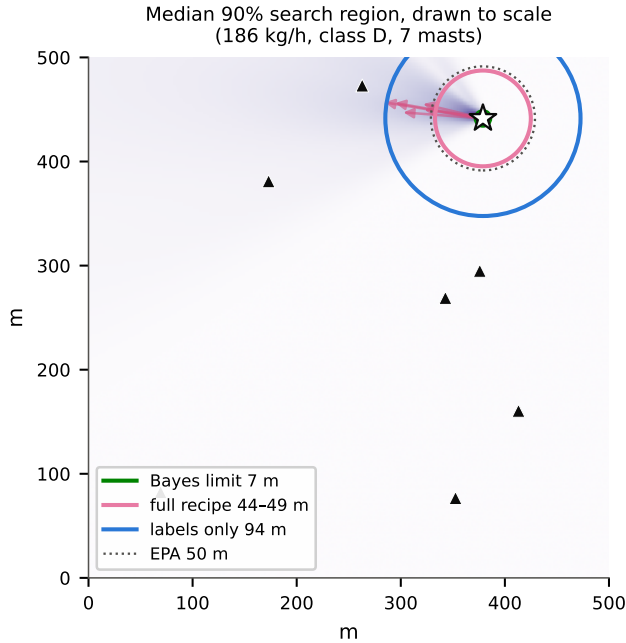


Fig. 1. The problem, to scale. A wind-swept methane plume (purple; meander arrows), seven masts (black), and the true source (star) on a 500 m site. The data support a 7 m search circle (Bayes limit, green); a standard calibrated localizer delivers 94 m (blue); the EPA actionability radius (50 m) sits between. Our recipe: 44–49 m (magenta). Calibration is not actionability.

safe tail-mass form. Region sizes are reported as equal-area circle radii.

III. DIAGNOSIS BY ELIMINATION

Calibrated $\neq$ actionable. A DeepSets localizer trained on one-hot labels (M0) is calibrated (coverage 0.89) with a median radius of 94 m — $13\times$ the Bayes limit in radius. The information was in the readings; the model could not extract it.

The teacher is not the bottleneck. Distilling the exact posterior requires tempering: the raw near-delta posterior fails outright (134 m, *worse than no physics*), while a 0.75-cell blur of it works (74 m). Is the blur the ceiling? Run the audit *on the tempered teacher itself*: it supports 11.7 m. The student sits $6\times$ above its own textbook — the failure is approximation, not supervision content.

Supervision cannot fix it. Annealing the temper toward the raw posterior makes things worse (87 m); a debiased Sinkhorn transport loss with ε -scaling also fails (243 m; implementation-scoped). Replacing the categorical head with a flow-based neural posterior estimator [5], [6] — the current SBI standard, given a $2.5\times$ budget, a widened head, and full-band conditioning after adversarial review of our own implementation — improves the labels-only baseline (62.8–71.2 m) but remains far from the limit, and we verified the spline head can express 4.4 m regions, so its blur is not head-structural.

Verdict. Localizing requires correlating each of 4,096 candidate cells’ predicted space–time signature against all read-

TABLE I
ONE RECIPE, THREE ARCHITECTURES. MEDIAN 90% CONFORMAL REGION RADIUS (M); RANGE OVER 3 SEEDS; COVERAGE 0.89–0.91 IN EVERY CELL. REFERENCES: BAYES LIMIT 7.0 M, TEACHER CEILING 11.7 M, EPA BAR 50 M.

Encoder	Labels	+Distill	+Maps (full)
DeepSets	94–98	73–74	48.1–49.3
GNN	55–58	49–52	43.6–44.3
Set Transformer	69–75	56–60	44.7–48.9

TABLE II
MODERNITY CONTROL: THE CURRENT SBI-STANDARD HEAD, SAME ENCODER, DATA, AND AUDIT; 3 SEEDS PER ROW.

Model (DeepSets encoder)	Radius (m)	Coverage
Flow-NPE (fairness-hardened)	62.8–71.2	0.90–0.91
Flow-NPE + maps + distillation	55.5–56.4	0.89–0.90
Ours (grid head, full recipe)	48.1–49.3	0.89

ings — per-cell matched filtering — which a pooled set representation cannot carry. No target and no loss teaches a network what its forward pass cannot compute.

IV. METHOD: PHYSICS-GUIDED POSTERIOR DISTILLATION

The forward model is known; the missing computation has a formula. The recipe (Fig. 2) hands it over and lets learning do the rest. For any base localizer: (1) compute per cell c the sufficient statistics $a_c = \sum g_{cy}$, $b_c = \sum g_c^2$ of the linear-Gaussian model (g_c the unit-rate response) and feed two 64×64 channels, the matched-filter statistic $z_c = a_c / (\sigma \sqrt{b_c})$ and $\log b_c$; (2) add a 3-layer, zero-initialized conv head ($\sim 10k$ parameters, 0.2% of the base) whose output is summed with the localizer’s logits; (3) train with cross-entropy to the tempered exact posterior; (4) wrap with unchanged split conformal. The maps add no information — they are computed from the network’s own inputs — only a computation; the teacher adds no inference cost; zero-initialization makes every comparison controlled by construction.

Why not just use the teacher? On this testbed one should — it is the measuring instrument, tractable by construction. In deployment it vanishes: realistic dispersion has no closed form and simulator inversion costs hours per event [8] where monitoring needs seconds. Moreover (§VI) the teacher *breaks* under realistic wind error while its students do not. And “the teacher is just part of the dataset” is our thesis, not an objection: every simulated training set already contains its own uncertainty labels; standard training throws them away.

V. RESULTS

Table I: every ingredient improves every architecture; all nine full-recipe seed-runs land under the 50 m bar; and the spread across architectures collapses (94-vs-56 without the recipe, 49-vs-44 with it) — supervision and inputs, not encoder design, were the binding constraint. Paired Wilcoxon on per-scenario log region sizes: $p < 10^{-200}$; point accuracy is essentially unchanged — what changes is the shape of uncertainty,

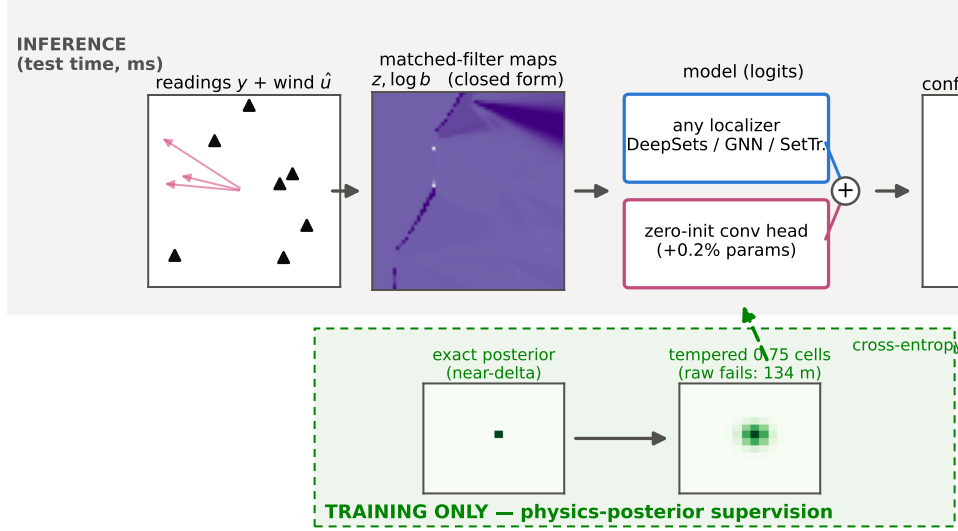


Fig. 2. The recipe, on real artifacts from one test scenario. (1) Observed mast readings and wind record. (2) Two closed-form physics maps computed in milliseconds from the same observations: the classical matched-filter statistic per candidate cell, and a sensitivity map (sufficient statistics of the forward model). (3) Any localizer, plus a zero-initialized convolutional head reading the maps: at step 0 the model is the unmodified baseline; training opens the physics channel only where it reduces loss. (4) Training targets are the exact Bayesian posterior of each scenario, tempered by a 0.75-cell blur (training only; no inference cost). (5) A split-conformal wrapper certifies 90% coverage; the median region shrinks from 94 m to 44–49 m against the EPA 50 m radius.

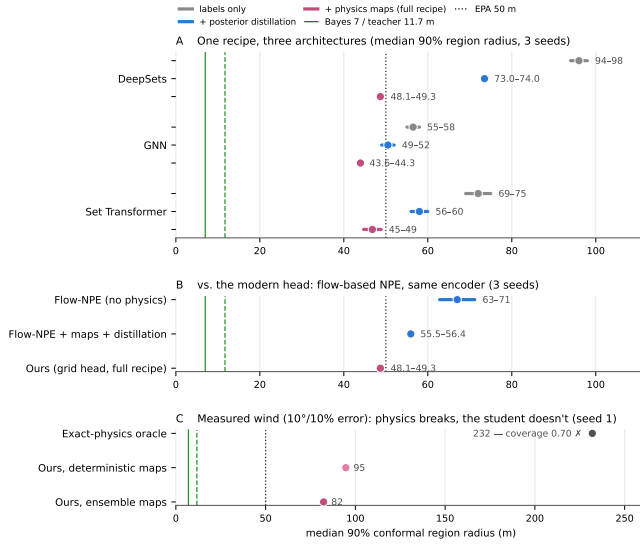


Fig. 3. All results on one axis. (A) The recipe across encoders; (B) the flow-head control; (C) measured-wind robustness. Vertical lines: Bayes limit 7 m (green), teacher ceiling 11.7 m (dashed), EPA 50 m (dotted). Whiskers span 3 seeds.

which is what region size measures. Table II: the modern head is a genuine lever (it beats plain distillation without any physics), our levers transfer onto it monotonically, and it still trails — the categorical head’s *spatial* physics injection (a conv over the maps) extracts more than the flow’s parametric conditioning at identical information. The recipe is compared against, not replaced by, the current standard.

TABLE III

WIND MEASURED WITH REALISTIC ERROR (10° DIRECTION, 10% SPEED): THE EXACT-PHYSICS PIPELINE COLLAPSES; THE STUDENTS DO NOT.

Pipeline	Coverage	Radius
Exact-physics oracle	0.70 ×	232 m
Ours, deterministic maps	0.91	94.7 m
Ours, ensemble maps	0.89	82.3 m

VI. ROBUSTNESS: MEASURED WIND

Real anemometers err. Under a 10°/10% observation error the “exact” posterior assigns *numerically zero* probability to the true source in 30% of scenarios — no threshold can cover a zero-mass cell, and the oracle’s conformal coverage collapses to 0.70 (Table III). The distilled student, trained through the same noise, remains calibrated: a softmax never says “impossible,” and training widens its uncertainty exactly where wind error demands. Making the physics inputs uncertainty-aware — *ensemble maps*, the mean and spread of the matched-filter maps over eight draws from the wind-error model — recovers 12 m ($p=10^{-122}$). Amortization is not merely cheap; it is robust to the misspecification that destroys its own teacher, and uncertainty in physics inputs must be modeled, not ignored.

VII. LIMITATIONS

CH4-T is an instrument, not a deployment: single steady sources (no zero-leak or multi-source scenarios), flat terrain without equipment wakes, known stability, quasi-static stepping, in-distribution marginal coverage. Rates sit in the super-emitter regime, above the $\leq 7 \text{ kg h}^{-1}$ range of standard METEC continuous-monitoring tests [9]; localization

granularity matches what such facilities score. The residual factor of ~ 6 to the 7 m limit is unexplained headroom; the transport-loss null is implementation-scoped; a wind-marginalized Bayes limit under measurement error is future work (the marginalization factorizes over timesteps, but reliable quadrature requires bearing-adaptive panels). Field claims require controlled-release validation; every comparison here is exact because all models share one verified simulator — fidelity limits transfer, not the tables.

VIII. CONCLUSION

Three numbers organize this work: the physics allows 7 m, regulators ask for 50 m, and a calibrated deep localizer delivers 94 m. The gap is not the sensors, the teacher, the loss, or the density estimator — it is a classical computation the network cannot rediscover. Handing it over, and distilling the uncertainty the dataset already contained, brings three different architectures under the actionability bar at unchanged cost, and survives the wind-measurement error that breaks exact physics inversion. The recipe is deliberately simple; the evidence standard — exact-posterior audits, blocking self-calibration gates, one-change-at-a-time controls with documented nulls — is the part we most hope transfers.

REFERENCES

- [1] UN Environment Programme (IMEO), “An eye on methane: Methane Alert and Response System annual report,” 2024.
- [2] U.S. EPA, “Standards of performance for GHG emissions: NSPS OOOOb/EG OOOOc and the Super-Emitter Program,” 89 FR 16820, 2024.
- [3] European Union, “Regulation (EU) 2024/1787 on methane emissions reductions in the energy sector,” OJ L, 2024.
- [4] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Found. Trends Mach. Learn.*, 16(4), 2023.
- [5] G. Papamakarios and I. Murray, “Fast ε -free inference of simulation models with Bayesian conditional density estimation,” *NeurIPS*, 2016.
- [6] C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios, “Neural spline flows,” *NeurIPS*, 2019.
- [7] J.-M. Lueckmann *et al.*, “Benchmarking simulation-based inference,” *AISTATS*, 2021.
- [8] T. Newman *et al.*, “Probabilistic inversion modeling of gas emissions: MCMC estimation of Gaussian plume parameters,” arXiv:2408.01298, 2024.
- [9] C. Ilonze *et al.*, “Assessing the performance of emerging and existing continuous monitoring solutions under a single-blind controlled testing protocol,” *Elementa*, 2025.